



The Effectiveness of Green Bonds on Renewable Energy Investment: The Moderating Role of Political Stability

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Abstract: In the context of conflicting empirical evidence regarding the effectiveness of green bonds on renewable energy investment, this study posits that the inconsistency in prior findings may stem from overlooking the role of the institutional environment. Accordingly, the study aims to examine the direct association between green bonds and renewable energy investment while analyzing the moderating role of political stability in this relationship. Using a panel dataset of 236 country-year observations from 16 emerging Asian economies over the 2010–2024 period, the study employs a Fixed Effects Model (FEM) with interaction terms and Driscoll-Kraay standard errors, complemented by robustness checks using System Generalized Method of Moments (GMM) estimation. The results reveal that green bonds are positively associated with renewable energy investment ($\beta_1 = 0.158$; $p < 0.01$). More importantly, the positive interaction coefficient ($\beta_3 = 0.092$; $p < 0.10$) suggests that political stability amplifies the association between green bonds and renewable energy investment. While this interaction effect is only marginally significant in the main specification, it gains further support from the System GMM estimation ($\beta_1 = 0.145$; $p < 0.05$; $\beta_3 = 0.105$; $p < 0.05$) and from subsample analysis, which reveals that the association between green bonds and renewable energy investment is statistically insignificant in politically unstable countries but strongly positive in stable ones. The study concludes that political stability appears to be an important enabling condition for realizing the potential of green finance in accelerating decarbonization, implying that green bond market development should go hand in hand with institutional reform and environmental governance strengthening.

Keywords: Asian economies; Emerging markets; Green bonds; Political stability; Renewable energy investment

1 Introduction

The urgent global imperative for climate change mitigation, enshrined in the Paris Agreement and the Sustainable Development Goals (SDGs), has driven an unprecedented effort toward energy transition. A critical challenge in this transition is the enormous capital requirement to shift from fossil fuels to renewable energy sources. This financial barrier is particularly acute for emerging economies, which simultaneously face rapidly growing energy demand and significant public finance constraints. The International Renewable Energy Agency (IRENA) and the International Energy Agency (IEA) have identified an annual investment gap of trillions of dollars, underscoring the insufficiency of public capital alone to finance this global effort [1, 2]. This context has spurred the search for innovative financial mechanisms capable of mobilizing private capital at the necessary scale.

To address this challenge, green finance has emerged as a potential complementary solution, with green bonds as its most prominent instrument. Defined as debt securities whose proceeds are committed exclusively to projects with environmental benefits [3], green bonds have experienced exponential growth, with cumulative market size surpassing several trillion dollars [4]. This growth has been particularly notable in Asia, a region pivotal to the global energy transition. The fundamental mechanism of green bonds is to signal the environmental standards of a project, thereby attracting sustainability-oriented investors [5, 6]. Prior research on green bonds has primarily focused on financial aspects. Specifically, these studies have documented the existence of a “greenium”—that is, green bonds typically carry lower yields than conventional bonds [7–9]. More recently, several studies have begun examining the broader role of green bonds in advancing sustainability objectives, including stimulating green investment and reducing carbon emissions [10, 11]. However, the empirical evidence remains inconsistent and often inconclusive,

with some studies finding positive effects [12], while others report insignificant effects [13, 14], suggesting that the link between green bond issuance and tangible environmental outcomes is more complex than initially assumed [15].

However, the effectiveness of any financial instrument cannot be examined in isolation; it depends profoundly on the institutional context of the host country. This study argues that the mixed results in the existing literature may stem from the fact that prior studies have overlooked (or underemphasized) a critical factor: the role of the political environment. Drawing on the foundations of New Institutional Economics, this study argues that robust institutions are essential for facilitating complex, long-term economic transactions by reducing transaction costs and protecting property rights [16]. Investments in renewable energy are quintessential examples of such transactions. They are characterized by high upfront capital costs, long payback periods, and heavy dependence on the stability and credibility of government support policies, such as feed-in tariffs (FITs) and tax incentives. This makes them particularly sensitive to political risk. Political instability—including the potential for government change, policy reversal, social unrest, or asset expropriation—creates a profound level of uncertainty that can deter even the most committed investors [17, 18].

This creates an important but largely unexplored disconnect in prior research. One stream of literature highlights the financial innovation of green bonds. Another stream, often independently, points to the adverse effects of political risk on sustainable investments. This disconnect raises a critical question with profound policy implications. Does the financial structure of green bonds function as an effective risk-mitigation mechanism that shields renewable energy investment from an unstable political environment? Or is the institutional context of the host country the ultimate determinant, neutralizing “green” signals in environments with high uncertainty? To date, the interaction between green financial mechanisms and political risk remains a significant theoretical gap. Failure to clarify this interaction may lead to inaccurate assessments of the true effectiveness of green bonds. It may also promote policy solutions ill-suited to the diverse political contexts of emerging economies.

Therefore, this study is conducted to fill this critical research gap. Rather than simply asking whether green bonds are effective, it poses a deeper and more policy-relevant question: under what institutional conditions is the effectiveness of green bonds optimized? To answer this question, the study systematically tests the moderating role of political stability in the relationship between green bond issuance and renewable energy investment, using a diverse panel dataset of 16 emerging Asian economies. The paper has two main objectives: (i) to test the direct effect of green bond market development on renewable energy investment in the study sample, and (ii) to examine whether political stability amplifies or diminishes this effect through formal interaction hypothesis testing.

This study makes three specific contributions to the existing body of knowledge. First, to the best of the author’s knowledge, this is among the first studies to formally model and test the interaction between green bond market development and political stability as determinants of renewable energy investment outcomes, thereby integrating the literature on green finance with the political economy of institutions. Although prior studies have examined green bonds and institutional quality separately, the explicit interaction between these two factors has not been tested. Second, unlike studies that estimate a single average effect of green bonds across heterogeneous samples, this study’s approach demonstrates that the effect is conditional, providing a more realistic and policy-relevant assessment of when and where green financial instruments are effective. Third, by focusing on emerging Asian economies—a region that is both central to the global energy transition and exhibits considerable diversity in political environments—the study generates context-specific evidence with direct applicability for policymakers and international development organizations. These findings carry not only academic significance but also practical implications for governments seeking to optimize the use of green finance as a tool for accelerating the energy transition.

2 Literature Review and Hypothesis Development

2.1 Green Bonds and Renewable Energy Investment

The theoretical basis for why green bonds may promote renewable energy investment rests primarily on signaling theory. By committing the raised capital to environmentally beneficial projects and subjecting themselves to external oversight, green bonds emit a credible signal about the environmental quality of a project, thereby reducing information asymmetry between the issuer and investors [3, 7]. This certification mechanism attracts sustainability-oriented investors, enabling green projects to reduce their cost of capital through the “greenium”—a cost-of-capital advantage over conventional debt instruments that has been widely documented [8, 19].

However, the link between the use of this financial instrument and actual increases in renewable energy investment has not been convincingly established, as empirical evidence remains contradictory. On the one hand, Kant [12] and Azhgaliyeva et al. [20] found a positive association between green bond issuance and clean energy deployment, arguing that green bonds serve not only as capital-raising tools but also as catalysts for broader environmental commitments. Flammer [21] further demonstrated that firms issuing green bonds subsequently increase their environmental expenditures relative to matched control firms, suggesting that green bonds genuinely create an “additionality” effect by stimulating new environmental investments. On the other hand, studies such as Fatica and Panzica [13] cautioned that many green bonds may finance projects that would have been undertaken regardless

of whether green bonds existed, raising concerns that green bonds do not truly generate additional environmental investments. Oktavio and Riyanti [14] also found that the cost-of-capital advantage in the Association of Southeast Asian Nations (ASEAN) does not necessarily translate into increased project investment.

This inconsistent evidence suggests that the relationship between green bonds and renewable energy investment is moderated by contextual factors that the existing literature has not adequately addressed. This observation underscores that investigating the role of the institutional environment is both warranted and necessary.

2.2 Political Stability, Institutional Quality, and Investment Decisions

A substantial body of literature, rooted in New Institutional Economics [16], affirmed that institutional quality is a fundamental determinant of investment behavior. Strong institutions reduce transaction costs, protect property rights, and enhance the credibility of contractual commitments—all of which are prerequisites for long-term, capital-intensive investments [22]. Renewable energy projects are a prime example: they require high upfront capital costs, payback periods of 15 to 25 years, and are closely dependent on the continuity of government support policies such as FITs and renewable portfolio standards [23].

Political instability poses a direct threat to the investment climate. Julio and Yook [17] showed that political uncertainty causes firms to delay investment, particularly in irreversible, asset-specific capital such as specialized facilities and equipment—assets that cannot be easily resold or repurposed. Rashid and Rashid [18] documented the multidimensional negative effects of political instability on economic activity. Specifically in the renewable energy sector, Aguirre and Ibikunle [24] demonstrated that policy and political risks are among the most significant barriers to mobilizing private finance for clean energy in developing countries. When governments change unexpectedly, FITs may be revised, power purchase agreements may be renegotiated, or regulatory frameworks may change entirely—rendering previously viable projects uneconomical [25].

Although extensive research exists on institutions and investment, as well as on green bonds and environmental outcomes, the link between these two domains has received inadequate attention. Specifically, very few studies examine how institutional factors (such as laws and policies) affect the effectiveness of green financial instruments. Most studies on green bonds implicitly assume that their impact is uniform across countries—an assumption that this study seeks to test.

2.3 Theoretical Framework and Hypothesis Development

Building on the theoretical foundations from the two domains analyzed above, this study argues that political stability plays a moderating role, either strengthening or weakening the impact of green bonds on renewable energy investment. This moderation operates through three main channels.

The first is the policy credibility channel. Renewable energy investment is particularly dependent on long-term government commitments. In a politically stable environment, these commitments are perceived as credible and durable, allowing investors to incorporate the financial benefits of green bond-financed projects into long-term return calculations with confidence. In an unstable environment, the perceived probability of policy reversal increases sharply, undermining the investment case regardless of the financing structure [17, 23].

The second is the risk premium channel. Political instability drives up the country risk premium demanded by investors. When this premium is sufficiently large, it overwhelms the modest cost-of-capital advantage that green bonds provide through the greenium. In such environments, the marginal benefit of green bond financing becomes economically insignificant relative to the overriding concern of political risk. Conversely, in stable environments where the baseline risk premium is low, the greenium and signaling benefits of green bonds become meaningful factors in capital allocation decisions [7, 24].

The third is the regulatory enforcement channel. The credibility of the environmental certification on green bonds depends partly on the effectiveness of the legal and regulatory system in the host country. In politically stable environments with strong governance, investors trust that the raised capital will be used for its intended purpose and that contractual obligations will be enforced. Conversely, weak governance environments may breed skepticism about the integrity of the “green” label itself, diminishing its signaling value [5, 25].

Based on this theoretical framework, the study formulates the following hypotheses:

H1: Green bond market development is positively associated with renewable energy investment in emerging Asian economies.

H2: Political stability positively moderates the relationship between green bond market development and renewable energy investment. Specifically, the positive effect of green bonds on renewable energy investment is stronger in countries with higher levels of political stability.

3 Methodology

3.1 Sample and Data Collection

The sample construction follows a systematic, multi-stage procedure designed to produce a dataset that is both

relevant to the research question and robust for econometric analysis. The final dataset is an unbalanced panel covering 16 emerging Asian economies over a 15-year period from 2010 to 2024. The selected countries are: Bangladesh, China, India, Indonesia, Jordan, Kazakhstan, Malaysia, Mongolia, Pakistan, the Philippines, South Korea, Sri Lanka, Thailand, Turkey, Uzbekistan, and Vietnam.

The initial population comprised 28 nations classified as emerging or developing economies in Asia by the International Monetary Fund (IMF) and the World Bank. The selection of the 2010–2024 timeframe is deliberate. This period uniquely captures the entire lifecycle of the green bond market in these regions, from its nascent stage (pre-2014) through its subsequent period of exponential growth. This temporal scope provides essential variation in the key independent variable, which is crucial for identifying its association with renewable energy investment.

To ensure data quality and the suitability of the sample for the regression models, a sequential set of exclusion criteria was applied:

(i) Data availability for green bonds (independent variable): Countries were first screened for any recorded green bond issuance using comprehensive data from the Climate Bonds Initiative (CBI) and Refinitiv Eikon. Countries with no green bond activity throughout the entire study period ($n = 3$) were excluded, as their inclusion would offer no information on the relationship of interest.

(ii) Data availability for renewable energy investment (dependent variable): Consistent data on annual renewable energy investment from International Renewable Energy Agency (IRENA) and the IEA were then required. Countries with significant data gaps (i.e., missing data for more than 50% of the observation years) were removed ($n = 4$) to prevent substantial loss of statistical power and potential sample selection bias.

(iii) Data availability for moderating and control variables: Finally, the availability of data was verified for the moderator (political stability, sourced from the World Bank’s Worldwide Governance Indicators (WGI)) and for the core macroeconomic controls (sourced from the World Development Indicators (WDI)). Countries with missing data for two or more key control variables across the majority of the period were excluded ($n = 5$), as their inclusion would risk omitted variable bias and compromise the reliability of the estimates. Table 1 summarizes this sequential screening process.

Table 1. Sample selection process

Stage	Criterion	Countries Excluded	Remaining Countries
Initial population	Emerging/developing Asian economies (IMF & World Bank classification)	–	28
Step 1	No recorded green bond issuance during 2010–2024	3	25
Step 2	Missing renewable energy investment data for >50% of observation years	4	21
Step 3	Missing data for ≥ 2 key control variables for a majority of the period	5	16
	Final sample		16 countries

The data filtering process culminated in a final unbalanced panel of 236 country-year observations. This sample size establishes a firm basis fully adequate for robust panel data analysis. All primary data were sourced from internationally recognized databases: IRENA, the World Bank (WDI and WGI), CBI, Refinitiv Eikon, and the U.S. Energy Information Administration (EIA). To mitigate the undue influence of extreme values, all continuous variables used in the models were winsorized at the 1st and 99th percentiles.

3.2 Variable Measurement

The selection and measurement of the variables are grounded in established theoretical frameworks and prior empirical literature. Table 2 summarizes the variables, their proxies, and data sources.

The measurement of the dependent variable, renewable energy investment (REI_GDP), and the independent variable, green bonds (GB_Stock_GDP), in this study has a specific rationale. Both are measured as ratios to GDP. This approach, following seminal studies in finance and investment (e.g., King and Levine [26]), is superior to using absolute values because it controls for economic size. It enables meaningful comparisons between large economies such as China and smaller ones such as Sri Lanka, focusing on investment intensity and financial activity relative to economic scale. This measurement directly follows Tolliver et al. [15] and Eyraud et al. [27], who have established it as a standard metric in the clean energy field.

For green bonds, the study uses the cumulative stock of bonds outstanding rather than annual issuance flows. This choice has a theoretical basis in the financial development literature, which distinguishes between market activity (flows) and market depth (stocks) as determinants of real economic outcomes [26, 28]. While annual issuance reflects

short-term market dynamism, the cumulative stock better represents the structural depth and long-term financial commitment of a country's green bond market. This measurement follows the methodological precedent established by Ehlers and Packer [5] and Banga [7] in their analyses of green bond market maturity, as well as the broader public debt literature, where the stock of outstanding debt is the standard measure for assessing financial market development. However, the study uses annual issuance as an alternative measure in robustness checks to ensure that results are not driven by this specific variable measurement.

Table 2. Definition and measurement of variables

Variable Role	Variable Name	Symbol	Measurement (Proxy)	Data Source
Dependent	Renewable energy investment	REI_GDP	The natural logarithm of the ratio of total annual investment in renewable energy to Gross Domestic Product (GDP)	IRENA, WDI
Independent	Green bonds	GB_Stock_GDP	The natural logarithm of the ratio of the cumulative stock of green bonds outstanding at year-end to GDP	CBI, Refinitiv, WDI
Moderating	Political stability	PS	The “Political Stability and Absence of Violence/Terrorism” index. The original scale ranges from approximately -2.5 (low stability) to +2.5 (high stability)	WGI
Control	Economic development	GDP_PC	The natural logarithm of GDP per capita (constant 2015 USD)	WDI
Control	Economic growth	GDP_Growth	The annual percentage growth rate of GDP	WDI
Control	Foreign direct investment	FDI	Net inflows of foreign direct investment as a percentage of GDP	WDI
Control	Trade openness	Trade	The sum of exports and imports of goods and services as a percentage of GDP	WDI
Control	Energy prices	Energy_Price	The natural logarithm of the annual average price of Brent crude oil (USD per barrel)	EIA, World Bank

Note: IRENA, International Renewable Energy Agency; WDI, World Development Indicators; CBI, Climate Bonds Initiative; WGI, Worldwide Governance Indicators; EIA, Energy Information Administration.

The political stability index from the WGI is a widely accepted proxy in the political economy and finance literature for measuring institutional quality and country risk (e.g., Aisen and Veiga [29]). This composite index captures perceptions of the likelihood of political instability and/or politically motivated violence, including terrorism, which is directly relevant to the long-term risks faced by large-scale energy projects. However, the study acknowledges that this index is perception-based and aggregates multiple dimensions of instability—from routine government changes to armed conflict—which may affect renewable energy investors heterogeneously. For example, the risk of abrupt policy reversal following a government change is conceptually distinct from the risk of armed conflict, yet both contribute to the composite score. The study adopts this index because it is the most comprehensive and widely used measure currently available for cross-country comparisons, and has been validated in numerous investment studies [17, 22, 29].

The control variables are selected based on standard practices in investment models. GDP_PC and GDP_Growth control for the level of development and economic dynamism. FDI accounts for external capital flows that may also finance renewable energy projects. Trade openness captures the degree to which a country is integrated with the global economy. Additionally, energy prices (proxied by oil prices) are included to account for a push effect: higher oil prices provide countries with greater incentive to transition to renewable energy.

3.3 Econometric Model

To test the hypotheses, the study specifies a panel data regression model that includes an interaction term to capture the moderating effect of political stability. The baseline model is specified as follows:

$$\ln(\text{REI_GDP})_{it} = \beta_0 + \beta_1 \ln(\text{GB_Stock_GDP})_{it} + \beta_2 \text{PS}_{it} + \beta_3 [\ln(\text{GB_Stock_GDP})_{it} \times \text{PS}_{it}] + \beta_4 \text{GDP_PC}_{it} + \beta_5 \text{GDP_Growth}_{it} + \beta_6 \text{FDI}_{it} + \beta_7 \text{Trade}_{it} + \beta_8 \ln(\text{Energy_Price})_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

where, i and t denote the country and year indices, respectively. The key coefficients of interest are β_1 , capturing the direct effect of green bond market depth on renewable energy investment; β_2 , capturing the direct effect of political stability; and most importantly, β_3 , the coefficient on the interaction term $[\ln(\text{GB_Stock_GDP})_{it} \times \text{PS}_{it}]$, which tests whether the effect of green bonds varies across different levels of political stability. A positive and statistically significant β_3 would provide direct evidence for H2. The vector of control variables includes

GDP_PC_{it} , GDP_Growth_{it} , FDI_{it} , $Trade_{it}$, and $\ln(Energy_Price)_{it}$. The terms α_i and γ_t represent country fixed effects and year fixed effects, respectively. Country fixed effects absorb all time-invariant unobserved heterogeneity (such as geography and legal origin), while year fixed effects control for common global shocks (such as the COVID-19 pandemic and commodity price fluctuations). ε_{it} is the idiosyncratic error term.

3.4 Analytical Strategy

First, the study conducts descriptive statistics and Pearson correlation analysis to preliminarily examine variable distributions and multicollinearity. The Hausman test is then used to select the appropriate panel estimation method.

A key concern in the empirical context of this study is endogeneity, which may arise from two sources. First, reverse causality is entirely plausible: countries that invest heavily in renewable energy may also proactively develop their green bond markets to finance these investments, creating a simultaneous relationship. Second, omitted variable bias may occur if unobserved time-varying factors (e.g., changes in government environmental ambitions) simultaneously drive both green bond market growth and renewable energy investment. To address this, the study incorporates country fixed effects, while year fixed effects control for common time shocks. To further address the potential for reverse causality and dynamic endogeneity, the study employs System Generalized Method of Moments (GMM) estimation (Blundell and Bond [30]) as a key robustness check, using lagged levels and differences of endogenous variables as instruments.

The study also acknowledges a potential temporal misalignment in the model specification. The independent variable (cumulative green bond stock) is a stock measure, while the dependent variable (annual renewable energy investment) is a flow measure. Because renewable energy investment tends to be lumpy and the lag between bond issuance and actual capital expenditure can vary, the contemporaneous relationship may not fully capture the dynamic linkage. The use of cumulative stock partially mitigates this concern, as it reflects the aggregate market depth that influences the investment environment over time rather than within a single year. Additionally, the robustness check using annual issuance flows provides a complementary perspective on this relationship.

To ensure reliable statistical inference in the Fixed Effects models (FEM), all standard errors are computed using the Driscoll-Kraay method, which simultaneously corrects for heteroskedasticity, autocorrelation, and cross-sectional dependence. Finally, the study conducts subsample analysis by splitting the data at the median value of political stability. While the fixed effects structure, Driscoll-Kraay standard errors, and System GMM robustness check help mitigate several sources of bias, the results should be interpreted as conditional associations rather than definitive causal effects. Accordingly, the language used throughout the results and discussion sections reflects this associational framing. All analyses are performed using Stata, version 17.

4 Results

4.1 Descriptive Statistics and Correlation Analysis

Table 3 presents descriptive statistics and the correlation matrix for all variables used in the analysis, based on the panel dataset of 236 country-year observations from 16 emerging economies in Asia over the 2010–2024 period.

Table 3. Descriptive statistics and correlation matrix

Panel A: Descriptive Statistics					
Variable	Observations	Mean	Std. Dev.	Min	Max
$\ln(REI_GDP)$	236	-5.452	1.681	-8.914	-2.103
$\ln(GB_Stock_GDP)$	236	-7.118	2.815	-12.540	-3.056
PS	236	-0.213	0.855	-2.380	1.150
GDP_PC	236	8.602	1.021	6.711	10.432
GDP_Growth	236	4.881	3.129	-5.760	11.950
FDI	236	2.754	3.607	-2.150	16.230
Trade	236	82.430	35.190	31.500	188.700
$\ln(Energy_Price)$	236	4.259	0.312	3.761	4.736
Panel B: Pearson Correlation Matrix					
	$\ln(REI_GDP)$	$\ln(GB_Stock_GDP)$	PS	GDP_PC	GDP_Growth
$\ln(REI_GDP)$	1.000				
$\ln(GB_Stock_GDP)$	0.478***	1.000			
PS	0.215**	0.189**	1.000		
GDP_PC	0.591***	0.512***	0.433***	1.000	
GDP_Growth	0.102	0.088	-0.157*	0.054	1.000
FDI	0.334***	0.241***	0.119	0.287***	0.198**
Trade	0.256***	0.301***	0.195**	0.355***	0.041
$\ln(Energy_Price)$	0.188**	0.095	-0.067	0.123	0.210**

Note: PS, political stability. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

The descriptive statistics reveal considerable variation, reflecting the fundamental economic realities of the study sample. The dependent variable, $\ln(\text{REI_GDP})$, has a mean of -5.452 (corresponding to a renewable energy investment-to-GDP ratio of approximately 0.43%) with a standard deviation of 1.681, indicating substantial heterogeneity in clean energy investment levels across countries and over time. This variation reflects the diverse stages of energy transition within the sample, ranging from early-stage markets such as Mongolia and Uzbekistan to more advanced deployers such as China and India.

The main independent variable, $\ln(\text{GB_Stock_GDP})$, exhibits even greater variation (standard deviation = 2.815), confirming the rapid but highly uneven development of this nascent market. This dispersion is plausible, as green bond markets in many sample countries were virtually nonexistent before 2015 and have grown exponentially since then. The political stability index ranges widely from -2.38 to 1.15, with a mean of -0.213. The negative mean indicates that the average country in the sample leans toward the unstable end of the global distribution, which is characteristic of emerging economies. This wide range—encompassing countries with relatively stable political environments such as South Korea and Malaysia to those with significant political turbulence such as Pakistan and Sri Lanka—provides the essential variation needed for a reliable test of the moderating role of this variable.

The correlation matrix in the lower panel of Table 3 provides preliminary insights into the relationships among variables. Notably, there is a positive and statistically significant correlation at the 1% level between $\ln(\text{GB_Stock_GDP})$ and $\ln(\text{REI_GDP})$ (coefficient = 0.478), providing initial evidence supporting the first research objective that green bond issuance is associated with higher levels of renewable energy investment. To examine multicollinearity, the study analyzes the correlation coefficients among independent variables and computes the Variance Inflation Factor (VIF). The results show that these correlation coefficients are all at low to moderate levels. Furthermore, the mean VIF of the model is 1.59, and no independent variable has a VIF exceeding the threshold of 2.5. This confirms that multicollinearity is not a serious concern, ensuring the reliability of the regression estimates in subsequent analyses.

4.2 Main Regression Results

Prior to analyzing the results, the study conducted model selection tests. The F-test is statistically significant at the 1% level, indicating that the FEM is more appropriate than the Pooled Ordinary Least Squares (OLS) model. More importantly, the Hausman test is also highly significant (p -value < 0.01), indicating that the FEM is the more appropriate and efficient choice over the Random Effects Model (REM) for controlling unobserved, time-invariant country characteristics. Accordingly, all main analyses are based on the FEM. To ensure robust estimation, all models employ Driscoll-Kraay standard errors, which simultaneously correct for heteroskedasticity, autocorrelation, and cross-sectional dependence in panel data. The main regression results are presented in Table 4.

The results from Model (2) provide initial evidence for the first research objective. The coefficient of $\ln(\text{GB_Stock_GDP})$ is 0.158 and statistically significant at the 1% level. This indicates that, without accounting for the role of the political environment, an increase in the scale of the green bond market is positively and significantly associated with renewable energy investment.

Further analysis focuses on Model (4), the most comprehensive model of the study. The results from this model yield important findings.

First, regarding the direct association with green bonds. In the model with the interaction term, the coefficient of $\ln(\text{GB_Stock_GDP})$ is 0.125. This coefficient represents the estimated marginal association between green bonds and renewable energy investment when the Political Stability index equals zero ($\text{PS} = 0$). Although still positive, this coefficient has decreased in both magnitude and significance compared to Model (2). This suggests that the association between green bonds and renewable energy investment is not constant but appears to depend on contextual factors.

Second, and central to the study, regarding the moderating role of political stability. The coefficient of the interaction term $\ln(\text{GB_Stock_GDP}) \times \text{PS}$ is 0.092 and statistically significant at the 10% level. While this level of significance warrants cautious interpretation, it provides suggestive evidence consistent with the second research objective. A positive interaction coefficient is consistent with the proposition that political stability amplifies the positive association between green bonds and renewable energy investment. In other words, the association between green bond market development and renewable energy investment appears to be stronger in countries with a more stable political environment.

To illustrate the economic magnitude, consider Model (2): the coefficient of 0.158 implies that a 10% increase in the green bond stock-to-GDP ratio is associated with an approximately 1.58% increase in the renewable energy investment-to-GDP ratio, holding other factors constant. In the full model (Model 4), this association becomes conditional on political stability. At the sample mean of PS (-0.21), the total estimated marginal association is $0.125 + 0.092 \times (-0.21) \approx 0.106$, suggesting that a 10% increase in the green bond ratio is associated with an approximately 1.06% increase in renewable energy investment. This magnitude is economically meaningful, suggesting that the association between green bonds and renewable energy investment is not trivial and that green bond market development may contribute to accelerating capital flows into clean energy beyond what would be expected from

general economic growth alone.

Table 4. The effect of green bonds on renewable energy investment: the moderating role of political stability

Dependent Variable	(1) ln(REI_GDP)	(2) ln(REI_GDP)	(3) ln(REI_GDP)	(4) ln(REI_GDP)
ln(GB_Stock_GDP)		0.158*** (0.041)	0.149*** (0.043)	0.125** (0.058)
PS			0.205** (0.092)	0.198** (0.095)
ln(GB_Stock_GDP) × PS				0.092* (0.049)
GDP_PC	0.612*** (0.155)	0.540*** (0.148)	0.521*** (0.151)	0.515*** (0.150)
GDP_Growth	0.028* (0.015)	0.025 (0.016)	0.027 (0.016)	0.026 (0.016)
FDI	0.045** (0.020)	0.039** (0.019)	0.038* (0.019)	0.037* (0.019)
Trade	0.005 (0.004)	0.002 (0.004)	0.001 (0.004)	0.001 (0.004)
ln(Energy_Price)	0.451** (0.198)	0.430** (0.190)	0.425** (0.188)	0.422** (0.187)
Constant	-11.85*** (1.450)	-11.21*** (1.398)	-11.05*** (1.402)	-10.98*** (1.395)
Country Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	236	236	236	236
Number of countries	16	16	16	16
Within R-squared	0.467	0.531	0.552	0.589
F-test (<i>p</i> -value)	0.000	0.000	0.000	0.000
Hausman test (<i>p</i> -value)	0.008	0.007	0.006	0.005

Note: All models are estimated using the Fixed Effects (within) estimator with Driscoll-Kraay standard errors (reported in parentheses), which correct for heteroskedasticity, autocorrelation, and cross-sectional dependence. The dependent variable in all columns is ln(REI_GDP), defined as the natural logarithm of the ratio of renewable energy investment to GDP. Model (1) includes only control variables. Model (2) adds the green bond variable ln(GB_Stock_GDP). Model (3) adds the political stability index (PS). Model (4) is the full model including the interaction term ln(GB_Stock_GDP) × PS. Within R-squared reports the proportion of within-country variation explained. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

To further clarify the economic significance of this moderating role, the study conducts marginal effect analysis of ln(GB_Stock_GDP) at different levels of political stability: low (25th percentile, PS = -0.78), sample mean (PS = -0.21), and high (75th percentile, PS = 0.36). The results are presented in Table 5.

Table 5. Marginal effect analysis of green bonds by level of political stability

Level of Political Stability (PS)	Marginal Effect of ln(GB_Stock_GDP) on ln(REI_GDP)	Standard Error	95% Confidence Interval
Low (PS = -0.78)	0.053	(0.071)	[-0.086, 0.192]
Average (PS = -0.21)	0.106*	(0.055)	[-0.002, 0.214]
High (PS = 0.36)	0.158***	(0.052)	[0.056, 0.260]

Note: Marginal effects are calculated from Model (4), Table 4. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

The marginal effect analysis shows that in countries with low political stability (PS = -0.78), the effect of green bonds on renewable energy investment is statistically insignificant. This indicates that in politically risky environments, the “green” signal from bonds is insufficient to translate into actual investment commitments. However, as the level of political stability increases, the effectiveness of green bonds becomes evident. At the average level of political stability, a 1% increase in the green bond stock-to-GDP ratio is associated with an approximately 0.106% increase in renewable energy investment-to-GDP. Notably, in countries with a highly stable political environment (PS = 0.36), this effect is amplified to 0.158% and is highly statistically significant. These results suggest that political and institutional stability may be an important enabling condition for realizing the potential of green financial instruments in channeling capital toward clean energy.

Finally, the control variables largely carry the expected signs. Economic development (GDP_PC) and foreign direct investment (FDI) have positive and significant effects on renewable energy investment, consistent with the theory that wealthier countries and those more open to foreign capital have greater resources for green projects.

Energy prices ($\ln(\text{Energy_Price})$) also carry a positive coefficient, indicating that higher fossil fuel prices create an economic incentive to shift toward renewable energy sources.

4.3 Robustness Checks

To confirm the robustness and reliability of the main results, the study conducts a series of robustness checks. These checks are designed to address three potential issues: (i) sensitivity of results to the measurement of the main independent variable; (ii) potential endogeneity; and (iii) consistency of results across different institutional environments. The results from these checks are compiled and presented in Table 6.

Table 6. Robustness checks

Method Sample Dependent Variable	(1) FEM Full Sample $\ln(\text{REI_GDP})$	(2) System GMM Full Sample $\ln(\text{REI_GDP})$	(3) FEM Low PS $\ln(\text{REI_GDP})$	(4) FEM High PS $\ln(\text{REI_GDP})$
$\ln(\text{GB_Flow_GDP})$	0.110* (0.061)			
$\ln(\text{GB_Flow_GDP}) \times \text{PS}$	0.085* (0.048)			
$\ln(\text{GB_Stock_GDP})$		0.145** (0.068)	0.065 (0.075)	0.215*** (0.060)
$\ln(\text{GB_Stock_GDP}) \times \text{PS}$		0.105** (0.051)		
PS	0.195** (0.090)	0.218** (0.102)		
GDP_PC	0.531*** (0.153)	0.498*** (0.165)	0.501*** (0.188)	0.542*** (0.179)
GDP_Growth	0.024 (0.016)	0.021 (0.018)	0.019 (0.020)	0.031* (0.018)
FDI	0.035* (0.019)	0.033 (0.022)	0.028 (0.025)	0.049** (0.021)
Trade	0.002 (0.004)	0.003 (0.005)	0.004 (0.006)	0.001 (0.005)
$\ln(\text{Energy_Price})$	0.415** (0.189)	0.395* (0.210)	0.435** (0.215)	0.408** (0.201)
Constant	-11.15*** (1.410)	-10.88*** (1.750)	-10.95*** (1.821)	-11.30*** (1.655)
Diagnostic tests				
Observations	236	205	107	129
Number of countries	16	16	16	16
Within R -squared	0.579	—	0.512	0.603
F -test (p -value)	0.000	—	0.000	0.000
AR(1) test (p -value)	—	0.021	—	—
AR(2) test (p -value)	—	0.255	—	—
Hansen test (p -value)	—	0.189	—	—

Note: GMM, Generalized Method of Moments; FEM, Fixed Effects Model. Column (1) replaces the green bond stock measure with the natural logarithm of annual green bond issuance to GDP ($\ln(\text{GB_Flow_GDP})$) as an alternative operationalization of green bond market activity. Column (2) employs the System GMM estimator to address potential endogeneity; the number of observations is reduced due to the use of lagged variables as instruments. AR(1) and AR(2) report p -values for the Arellano-Bond tests for first- and second-order serial correlation in the differenced residuals. The Hansen test reports the p -value for the test of overidentifying restrictions (instrument validity). Columns (3) and (4) present results from subsample regressions: the “Low PS” sample includes observations where the Political Stability index is at or below the sample median, and the “High PS” sample includes observations above the median. FEM models use Driscoll-Kraay standard errors; the GMM model uses robust standard errors. All values in parentheses are standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

First, the study examines whether the results are sensitive to the measurement of the main independent variable. The main model uses the cumulative green bond stock to measure market depth. In this check, it is replaced with annual issuance flows, a measure of short-term market activity. The results in Column (1) of Table 6 show that the coefficient of the interaction term, $\ln(\text{GB_Flow_GDP}) \times \text{PS}$, remains positive (0.085) and statistically significant at the 10% level. This demonstrates that the positive moderating role of political stability is maintained even when using an alternative measure of green bond market activity. This result confirms that the study’s main conclusion is not contingent on a specific definition of the green bond variable.

Second, to address endogeneity concerns—particularly the possibility that countries with strong renewable energy ambitions may proactively develop their green bond markets, or that an unobserved shock may simultaneously drive

both variables—the study employs System GMM estimation. This method uses lagged levels and differences of endogenous variables as instruments, allowing consistent estimation even in the presence of reverse causality and dynamic panel bias. The results in Column (2) show that the coefficients of both $\ln(\text{GB_Stock_GDP})$ (0.145; $p < 0.05$) and the interaction term $\ln(\text{GB_Stock_GDP}) \times \text{PS}$ (0.105; $p < 0.05$) maintain their positive signs and statistical significance, confirming the direction and substantive interpretation of the main findings.

Importantly, the validity of the GMM approach depends on instrument quality, which is verified through standard diagnostic tests. The Arellano-Bond test for serial correlation shows statistically significant first-order autocorrelation in the differenced residuals (AR(1) p -value = 0.021), which is expected and required by design, as the GMM transformation induces first-order correlation. More importantly, the second-order autocorrelation test is statistically insignificant (AR(2) p -value = 0.255), confirming that the untransformed errors are free from serial correlation—a necessary condition for the validity of lagged instruments. The Hansen test for overidentifying restrictions yields a p -value of 0.189, failing to reject the null hypothesis that the instruments are jointly valid and exogenous. This p -value falls within the acceptable range (above 0.10 but not suspiciously close to 1.00), suggesting that the model is neither misspecified nor overfitting. These diagnostic tests provide evidence that the GMM estimates are reliable, thereby reinforcing the main finding that the positive effect of green bonds on renewable energy investment is moderated by political stability.

Finally, the study performs subsample analysis to provide a more intuitive verification of the second research objective. The study divides the sample into two groups based on the median value of the political stability index: a “Low political stability” group and a “High political stability” group. The baseline model (without the interaction term) is then re-estimated on each subsample. The results in Columns (3) and (4) reveal a stark contrast. In the group of countries with low political stability (Column 3), the effect of $\ln(\text{GB_Stock_GDP})$ on renewable energy investment is statistically insignificant (coefficient = 0.065). Conversely, in the group of countries with a highly stable political environment (Column 4), the coefficient of $\ln(\text{GB_Stock_GDP})$ is not only substantially larger (0.215) but also statistically significant at the 1% level. This stark contrast provides strong nonparametric evidence supporting the main finding of the study: the association between green bonds and renewable energy investment depends systematically on the quality of the political and institutional environment of the host country.

5 Discussion

5.1 Interpretation of Key Findings

The results of this study support both hypotheses. First, the finding of a positive and statistically significant relationship between green bond market depth and renewable energy investment ($\beta_1 = 0.158$; $p < 0.01$ in Model 2) provides empirical evidence supporting H1, consistent with signaling theory in the context of green finance. This result aligns with Kant [12] and Flammer [21], suggesting that green bond issuance is associated not only with capital mobilization but also with a certification mechanism that may reduce information asymmetry and attract sustainability-oriented capital. The “green label” may help renewable energy projects differentiate themselves in competitive capital markets, potentially facilitating access to broader sources of capital and lowering financing barriers for clean energy projects.

The positive interaction coefficient ($\beta_3 = 0.092$; $p < 0.10$ in Model 4) provides suggestive evidence consistent with H2: the association between green bonds and renewable energy investment appears to be conditional on the political environment. Given that this coefficient is significant only at the 10% level in the main specification, the finding should be interpreted with appropriate caution. However, the consistency of this pattern across multiple analytical approaches lends additional credibility to the result. The marginal effect analysis (Table 5) and subsample regressions (Table 6, Columns 3–4) sharpen this picture. In politically unstable countries, the estimated association between green bonds and renewable energy investment is statistically insignificant (0.053; $p > 0.10$), suggesting that in environments where political risk is high, the “green signal” of bonds may be insufficient to influence investment behavior. Conversely, in countries with high political stability (PS at the 75th percentile), the estimated marginal association is considerably larger and highly statistically significant (0.158; $p < 0.01$). The subsample results reinforce this pattern: the green bond coefficient is statistically insignificant in the low-stability subsample (0.065) but large and significant in the high-stability subsample (0.215; $p < 0.01$). Moreover, the System GMM estimation, which addresses endogeneity concerns, yields a stronger interaction coefficient ($\beta_3 = 0.105$; $p < 0.05$), providing further support for the moderating role of political stability.

These findings may help reconcile the contradictory results in the prior literature. Studies that found no significant association between green bonds and environmental outcomes (e.g., Fatica and Panzica [13]; Oktavio and Riyanti [14]) may have analyzed samples dominated by weak institutional environments, where political risk offsets the green signal. Conversely, studies reporting positive associations (e.g., Kant [12]; Flammer [21]) may have relied on samples with stronger institutional foundations. The results of this study suggest that both sets of findings may be valid—the key may lie in the institutional context in which green bonds operate.

The observed pattern is consistent with the three channels outlined in the theoretical framework. Through the

policy credibility channel, stable governments ensure that critical policies (such as FITs, tax incentives, and energy targets) are sustained over the long term and are not altered after each election. This makes long-term cash flow projections for projects more reliable in the eyes of investors. Through the risk premium channel, when politics is stable, investment risk remains low. Only then does the modest financial benefit from green bonds (the greenium) become sufficiently attractive to influence investor decisions. Through the regulatory enforcement channel, a stable governance environment ensures that green bond certification carries genuine signaling value, as investors trust that the capital will be deployed for its intended purpose.

From an environmental policy perspective, these findings carry important implications for climate mitigation strategy design and implementation. Renewable energy investment is not merely a financial outcome; it is the primary mechanism through which countries reduce greenhouse gas emissions from the energy sector—the largest contributor to global warming. When green bonds fail to translate into actual investment in politically unstable environments, the consequence extends beyond financial inefficiency to a tangible delay in decarbonization and a widening gap between national emissions reduction pledges and on-the-ground progress. Countries that adopt green bond frameworks as part of their Nationally Determined Contributions (NDCs) under the Paris Agreement therefore need to recognize that the effectiveness of these financial instruments is embedded within the broader environmental governance architecture. A green bond program implemented without parallel efforts to ensure the continuity of energy transition planning, strengthen climate policy implementation frameworks, and maintain political stability is unlikely to deliver the expected environmental outcomes. This underscores the need for an integrated approach to environmental governance, in which financial innovation and institutional reform are treated as complementary pillars of a coherent decarbonization strategy.

5.2 Contributions and Policy Implications

Theoretically, this study integrates two largely independent streams of scholarship—green financial innovation and the political economy of institutions. By providing evidence that institutional quality may moderate, rather than merely accompany, the association between green bonds and renewable energy investment, the study challenges the implicit assumption that the impact of green bonds is uniform across contexts and proposes a conditional framework that better reflects the heterogeneous reality of emerging economies.

The practical and policy implications are substantial and can be articulated at three levels.

For national policymakers in emerging Asian economies, the findings suggest that developing the green bond market and strengthening institutional governance should be pursued as parallel, mutually reinforcing objectives in support of national climate targets and energy transition goals. Specifically, governments should: (i) codify renewable energy support policies (such as FITs and tax credits) into legislation, rather than issuing them solely through sub-legislative instruments (such as decrees or executive orders), to prevent easy reversal when governments change; (ii) establish independent energy regulatory agencies to insulate policy implementation from short-term political pressures; and (iii) adopt internationally recognized green bond standards and verification procedures (such as Climate Bond Standards) to enhance the credibility of domestic green bond markets.

For international financial institutions such as the World Bank, the Asian Development Bank (ADB), and bilateral development agencies, the findings suggest that technical assistance programs for green bond market development should be coupled with governance support. Specifically, development banks could structure conditional green bond guarantee instruments linked to institutional benchmarks—for example, tying partial credit guarantees to the maintenance of a stable renewable energy policy framework. Such conditionality would simultaneously mitigate political risk for investors and create incentives for institutional reform.

For private investors and asset managers, this study provides a quantitative analytical framework for integrating political risk into green asset allocation decisions. The marginal effect analysis demonstrates that the realized association between green bonds and renewable energy investment is substantially stronger in politically stable environments, providing an empirical basis for constructing risk-adjusted portfolios in emerging green bond markets.

5.3 Limitations and Future Research Directions

Although this study achieves its stated objectives, several limitations should be acknowledged. First, the Political Stability index from the WGI is a composite, perception-based measure that aggregates multiple dimensions of instability—from routine government changes to violent conflict—which may affect renewable energy investors differently. As discussed in the methodology section, the study adopts this index because it is the most widely validated cross-country institutional measure currently available, but future research could disaggregate the components of political risk to determine which specific dimension (e.g., policy reversal risk versus conflict risk) has the greatest influence on green bond effectiveness.

Second, the study focuses on a single institutional dimension. The broader institutional environment encompasses other factors such as regulatory quality, rule of law, and control of corruption, each of which may independently

moderate the green bond–investment relationship. Extending the analysis to incorporate multiple institutional dimensions, or constructing a composite institutional index, would provide a more comprehensive picture.

Third, although the sample of 16 emerging Asian economies is diverse, it treats “emerging Asia” as a homogeneous analytical category. These countries differ significantly in financial development, capital market openness, and institutional histories. While country fixed effects absorb time-invariant heterogeneity, the moderating effect of political stability may itself vary across subgroups (e.g., middle-income versus low-income countries, or financially open versus closed economies). Future research could explore this heterogeneity through additional subsample analyses.

Fourth, the use of a cumulative stock measure for the independent variable and an annual flow measure for the dependent variable creates a potential temporal misalignment, as discussed in the analytical strategy section. Although the robustness check with annual issuance flows partially addresses this issue, future research could employ lagged dependent variables or multi-year investment averages to better capture the dynamic relationship between green bond market development and renewable energy capital formation.

Finally, the analysis is conducted at the country level. Future research could undertake firm-level or project-level analyses to test whether green bond issuers actually increase their capital expenditure on renewable energy, thereby providing more direct evidence of the additionality of this instrument.

6 Conclusion

This study contributes to the debate on the effectiveness of green bonds by investigating the moderating role of political stability, using a panel dataset of 16 emerging Asian economies over the 2010–2024 period. The empirical analysis yields two key findings. First, green bond market development is positively associated with renewable energy investment, consistent with the signaling and financing functions of this instrument. Second, and more importantly, this association does not appear to be universal but rather conditional on the institutional environment of the host country. The positive association between green bonds and renewable energy investment is amplified in politically stable countries but becomes statistically insignificant in environments with high political risk—a pattern that is consistent across alternative variable measurements, System GMM estimation, and subsample analysis. While the interaction effect is only marginally significant in the main fixed effects specification, its consistency across multiple robustness checks and its strengthening in the GMM framework lend credibility to the moderating role of political stability.

The core implication is that, while green bonds are a financial instrument with considerable potential for financing the energy transition, they are unlikely to function as a standalone solution. Their effectiveness appears to depend on the institutional foundations upon which they operate. For emerging economies seeking to leverage green bonds to achieve renewable energy targets, reduce carbon emissions, and fulfill Paris Agreement commitments, this means that market development and governance reform should be pursued in tandem. Policymakers should embed renewable energy support mechanisms within durable legal frameworks that are resilient to political transitions, while international organizations should design green finance support programs that integrate both institutional strengthening and environmental governance capacity building. For investors, the institutional context of the host country should be a key factor in green asset allocation decisions. Only when the “green” signal of bonds is supported by a stable and predictable political environment is its potential to accelerate decarbonization and advance the energy transition likely to be fully realized.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflict of interest.

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